
AS Biology

7401/2 - Paper 2

Mark scheme

June 2018

Version/Stage: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Marking Guidance	Mark	Comments
01.1	B; A; E;	3	
01.2	1. (Many mitochondria) release energy / ATP for movement of vesicles / synthesis of protein / active transport; 2. (Many Golgi) vesicles transport protein / glycoprotein / milk to cell membrane / out of cell;	2	Must include function of organelle and use in context of milk production. Ignore reference to lipid / triglyceride 1. Reject reference to mitochondria undergoing anaerobic respiration 1. Reject “produce energy”. 1. Reject “energy for respiration” 2. Accept exocytosis as transport and release 2. Ignore references to protein synthesis

Question	Marking Guidance	Mark	Comments
02.1	1. Condensation (reaction) / loss of water; 2. Between amine / NH_2 and carboxyl / COOH ;	2	Accept each marking point if shown clearly in diagram. 2. Accept between amino (group) and carboxylic / acid (group)
02.2	1. Hydrogen bonds; 2. Between NH (group of one amino acid) and C=O (group); OR Forming β pleated sheets / α helix;	2	Accept as a diagram 1. Reject $\text{N} - - - \text{C}$ / ionic / disulfide bridge / peptide bond
02.3	1. Different sequence of amino acids OR Different primary structure; 2. Forms ionic / hydrogen / disulfide bonds in different places;	2	1. If candidate assumes proteins are the same, accept effect of different pH/ temperature

Question	Marking Guidance	Mark	Comments
03.1	As size increases, ratio (of surface area to volume) decreases;	1	Accept converse. Comparison required, e.g., <u>smaller</u> organisms have a <u>larger</u> ratio
03.2	Two marks for correct answer in range of 1.75 to 1.76032;;	2	Accept for 1 mark, incorrect answer using radius 0.87 / 0.88 / 0.880 / 0.8802 / 0.88015; OR Accept for 1 mark, incorrect answer with correct rearranged equation, e.g., Radius = $\sqrt{(\text{surface area} \div 4\pi)}$ OR $= \sqrt{9.73 \div 12.56}$ OR $= \sqrt{0.77} / \sqrt{0.774} / \sqrt{0.775}$ OR $r^2 = \text{surface area} \div 4 \pi$ OR $r^2 = 9.73 \div 12.56$ OR $r^2 = 0.77 / 0.774 / 0.775$
03.3	(Measures) small uptake / amount / quantity / volume / concentration / rate (of oxygen uptake); OR Avoids use of powers of ten / standard form / many decimal places;	1	Ignore weight / accuracy

03.4	More accurate / less error (in measuring mass); OR Causes less distress / damage to animal (to measure mass); OR Easier / quicker (to find mass) because irregular shapes; OR Fewer measurements / calculations;	1	Ignore references to human error Accept converse if reference made to volume Reject if comparison is made with surface area.
Question	Marking Guidance	Mark	Comments
03.5	(Oxygen used in) respiration, which provides energy / ATP; OR (Oxygen is used in) respiration, which is a metabolic process / chemical reaction;	1	Reject produces energy Reject references to anaerobic respiration
03.6	1. No information about egg; 2. So cannot compare all stages (in Table 2); 3. No statistical information / test / t-test / comparison of standard deviations; OR No measure of significant differences;	3	2. Idea of comparing all three stages needed 3. Reject statements that “results” are not significant 3. Reject references to chi squared or correlation coefficient

Question	Marking Guidance	Mark	Comments
04.1	1. (Movement) down a gradient / from high concentration to low concentration; 2. Passive / not active processes; OR Do not use energy from respiration / from ATP / from metabolism; OR Use energy from the solution;	2	1. Ignore along / across gradient 1. Reject movement from gradient to gradient 2. Reject do not use energy unqualified
04.2	1. Movement through carrier proteins; OR Facilitated diffusion; Between A and B 2. Rate of uptake proportional to (external) concentration; Between C and D 3. All channel / carrier proteins in use / saturated / limiting;	3	Accept MP1 in either section 1. Ignore co-transport / active transport 1. Accept channel proteins 2. Accept description of proportional 3. Accept used up 3. Accept transport proteins
04.3	1. Rate of uptake is proportional / does not level off (so diffusion occurring); 2. (Lipid-soluble molecules) diffuse through / are soluble in phospholipid (bilayer);	2	1. Accept as one increases the other increases

Question	Marking Guidance	Mark	Comments
05.1	1. Where dividing cells are found / mitosis occurs; OR No dividing cells / mitosis in tissue further away / more than 5mm from tip; OR To get (soft) tissue that will squash; OR Length that will fit under cover slip; 2. Single / thin layer of cells / spread out cells so light passes through (making cells / nuclei visible);	2	1. Accept most dividing cells 2. Accept thin layer of tissue 2. Ignore to see cells clearly
05.2	3.57 / 3.6 / 3.7 / 3.71 / 3.8 (%);;	2	If the answer includes additional decimal places, award the marks if it would round to a correct answer There are 3 cells in anaphase Accept for 1 mark, 101.25 / 101 (students estimate in minutes) OR 3.75 (difference between scientist estimate and student's estimate in minutes) Ignore plus or minus signs
05.3	Cytokinesis;	1	

05.4	Description; Explanation; E.g, 1. Examine large number of fields of view / many cells; 2. To ensure representative sample; OR 3. Repeat count; 4. To ensure figures are correct; OR 5. Method to deal with part cells shown at edge /count only whole cells; 6. To standardise counting;	2 max	Mark as pairs only 1. Accept large number / 20 or more for many 2. Accept typical / reliable
05.5	1. Stops anaphase / cell division / mitosis; 2. (By) stopping / disrupting / spindle fibres forming / attaching / pulling; 3. Preventing separation of (sister) chromatids; 4. (So) no new cells added (to root tip);	3 max	1. Accept prevents telophase / cytokinesis 2. Ignore affects anaphase 3. Ignore chromosomes separate / split 3. Accept chromatids split

Question	Marking Guidance	Mark	Comments
06.1	- Used to compare effect of other treatments / as a baseline; - Shows / Measures effect of substance (X); OR Accounts for effect of substances produced naturally;	2	Accept for 2 marks, substance (X) and not agar / block / water that caused the difference in the number of roots. - Do not accept unqualified reference to “compare results”. - Accept measures effect of independent variable
06.2	(D shows) substance (X) is not required for (some) root growth / production of roots; OR Substances (already) present in stem cause (some) root growth; - Substance X moves through plant; (E shows) substance (X) causes / increases / doubles number of roots / root growth;	3	Accept X moves through stem / phloem
06.3	In support of mass flow hypothesis (F shows) phloem is involved; (G shows) respiration / active transport is involved (in flow / movement); - Because 4 °C / cooling reduces / slows / stops flow / movement; - The agar block is the source; - Roots are the sink; Against the mass flow hypothesis - No bulge above ringing (in F); - No (role for) osmosis / hydrostatic pressure / water movement; - Movement could be due to gravity; - Roots still grow without (intact/functioning) phloem; - No leaves / sugars / photosynthesis to act as a source;	4 max	Each point must be clearly made in the context of support or against. Ignore sugar / sucrose 3 max for “support” and 3 max for “against” Accept no turgor pressure

Question	Marking Guidance	Mark	Comments
07.1	1. Hydrolysis (of); 2. (Large / insoluble substances) to small(er) / soluble substances;	2	Ignored named examples Accept polymer to monomer
07.2	1. Active sites are different shapes; 2. So different enzyme-substrate complexes (are formed); OR So complementary to different parts of cellulose / substrate;	2	
07.3	2×10^{-3} / 2.0×10^{-3} / 2.01×10^{-3} ;;	2	If the answer includes additional decimal places, award the marks if it would round to a correct answer Accept for 1 mark, correct answer not in standard form 0.002 / 0.00201 / 0.002014; OR Correct calculation using incorrect figure from table (9.2) 0.003 / 0.0031 / 0.00319 / 3×10^{-3} / 3.0×10^{-3} / 3.19×10^{-3} / 3.2×10^{-3} OR Correct calculation with answer expressed as g hr^{-1} , 0.12 / 0.121 / 1.2×10^{-1}
07.4	1. Endocellulase create more ends / increases surface area; 2. For exocellulase to act on / hydrolyse / digest;	2	

07.5	$\frac{\text{Initial Mass} - \text{Final Mass}}{\text{Initial Mass}} \times 100$	1	Accept ((final mass – initial mass) ÷ initial mass) x100 OR ((change in mass) ÷ initial mass) x100 OR $100 - \left(\frac{\text{final mass}}{\text{initial mass}} \times 100 \right)$ OR $(1 - (F/I)) \times 100$ OR $((I - F) \div I) \times 100 / ((F - I) \div I) \times 100$ OR $((15 - \text{final mass}) \div 15) \times 100 / ((\text{final mass} - 15) \div 15) \times 100$
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Question	Marking Guidance	Mark	Comments
08.1	1. Add 1 part (bacteria) culture to 9 parts (sterile) liquid (to make 10^{-1} dilution); 2. Mix (well); 3. Repeat using 9 parts fresh (sterile) liquid and 1 part of 10^{-1} and 10^{-2} dilutions to make 10^{-3} dilution; OR Add 1 part 10^{-1} (suspension) to 99 parts (sterile) liquid (to make 10^{-3} dilution);	3	1. and 3. Accept water / nutrient / broth for liquid 2. Accept stir 3. Reject 1 part (undiluted) culture added to 999 parts liquid
08.2	$3.75 \times 10^9 / 3\,750\,000\,000$;;	2	Accept for 1 mark: $3750\,000 / 3.75 \times 10^6$ (cells per mm^3) OR 3.75×10^{12} (wrong volume conversion) OR 3750 (cells per mm^3 of diluted culture) OR Evidence of using correct dilution conversion and correct volume conversion, i.e., $\times 1000$ and $\times 1000$
08.3	1. Count unlikely to be accurate / repeatable / reproducible / reliable; 2. Because too many cells; OR Because cells overlapping / not spread out;	2	

Question	Marking Guidance	Mark	Comments
08.4	1. Tetracycline used more often / in higher doses; 2. Resistant bacteria more likely to (survive and reproduce and) pass on allele/gene for (tetracycline) resistance; OR 3. More / higher frequency of mutations (for tetracycline resistance); 4. (so) gene passed on to more bacteria; OR 5. Tetracycline used over longer time period; 6. More time for (chance) mutation to occur / for selection to occur;	2	Ignore reference to resistant animals Ignore reference to immunity 3. Reject reference to mutation being caused by use of antibiotic
08.5	No selection against resistant bacteria / resistance gene/allele; OR Bacteria pass on (resistance) gene / allele when they reproduce; OR Bacteria resistant to tetracycline are passed on from one generation of farm animals to the next (probably via faeces); OR Environment does not change, so stabilising selection occurs;	1	Accept no selection to get rid of it Reject reference to mitosis or immunity

Question	Marking Guidance	Mark	Comments
09.1	Comparisons - Nucleotide structure is identical; - Nucleotides joined by phosphodiester bond; OR Deoxyribose joined to phosphate (in sugar, phosphate backbone); - DNA in mitochondria / chloroplasts same / similar (structure) to DNA in prokaryotes; Contrasts - Eukaryotic DNA is longer; - Eukaryotic DNA contain introns, prokaryotic DNA does not; - Eukaryotic DNA is linear, prokaryotic DNA is circular; - Eukaryotic DNA is associated with / bound to protein / histones, prokaryotic DNA is not;	5 max	- Accept labelled diagram or description of nucleotide as phosphate, deoxyribose and base - Accept shorter than nuclear DNA/is circular not linear/is not associated with protein/histones unlike nuclear DNA;
09.2	- Mutations change base / nucleotide (sequence); (Causing) change in amino acid sequence; - Mutations build up over time; - More mutations / more differences (in amino acid / base / nucleotide sequence / primary structure) between distantly related species; OR Few(er) mutations / differences (in amino acid / base / nucleotide sequence / primary structure) in closely related species; - Distantly related species have earlier common ancestor; OR Closely related species have recent common ancestor;	5	Accept “order” for “sequence” Reject if mutation in amino acid If neither MP4 or MP5 accept for 1 mark, idea of more mutations /differences as evidence of earlier common ancestor OR converse